

Deterministic Cosmology and Quantum Geometry: An Analysis of Unified Source Theory (UST) v5

Niyazi ÖCAL

Independent Researcher

Patent: TR 2026/003258 — Zenodo: DOI 10.5281/zenodo.19435766

April 2026

Abstract

Modern physics faces a profound crisis characterized by the ontological disconnect between quantum mechanics and general relativity, and the cosmological constant problem on the order of 10^{120} . This study presents the Unified Source Theory (UST) v5 model, built upon a universal dimensionless constant, $N_{s,q} \approx 0.63354460$. UST defines the universe as a dual-channel information architecture consisting of an active channel (Channel Q) and a frozen channel (Channel C). The proposed model resolves the Hubble tension with a 0.18% deviation, derives dark energy as the gravitational background pressure of Channel C, and explains the Big Bang as a fully conservative "Channel Inversion" phase transition. The validity of the theory depends on the precise falsification (falsifiability) barrier defined through 2028 ESA Euclid and CMB-S4 data.

1 Introduction

The current state of physical science is stalled by the cosmological constant problem defined by Steven Weinberg and the 5-sigma Hubble tension between local measurements and CMB data. Quantum error correction protocols standardized by Nielsen and Chuang require a massive "ancilla" (auxiliary) qubit load, which hinders hardware scalability. Furthermore, the Conformal Cyclic Cosmology (CCC) proposed by Roger Penrose (Oxford) requires parameters that have not yet been empirically verified, such as the vanishing of Weyl curvature.

Unified Source Theory (UST) resolves these crises without "fine-tuning" by using the $N_{s,q}$ constant derived analytically from the absolute limits of space-time geometry. This constant is obtained through the maximization of the G_{tt} component of the Einstein tensor and the addition of the Seeley-DeWitt $-\alpha/17$ quantum vacuum correction. The UST architecture views the universe as a unitary evolution of information rising from a pure potential called "Omnium". Information circulates between the active sector (Channel Q) and the frozen gravitational background (Channel C). This structure redefines decoherence not as noise, but as a deterministic sealing process into Channel C.

2 Section 2: Analytical Derivation of the $N_{s,q}$ Constant and Seeley-DeWitt (a_2) Correction

Unified Source Theory (UST) maintains that physical reality arises from the absolute limits of space-time geometry rather than randomly selected parameters. In a static and spherically symmetric metric structure, the G_{tt} component of the Einstein tensor, representing the geometric energy density of the system, is defined as follows:

$$G_{tt} = A(1 - A) + rA' \quad (1)$$

At the Omnium horizon ($r = r_s$), under the condition that the geometric contribution is maximized ($A' = 0$), the equilibrium state occurs at the $A = 1/2$ point. When the horizon ratio is defined in the UST framework as $A = (1 - N_{s,q})(2 - N_{s,q})$, the resulting quadratic maximization equation is:

$$N_{s,q}^2 - 3N_{s,q} + \frac{3}{2} = 0 \quad (2)$$

The physically meaningful root of this equation is determined as the pure geometric root of the universe ($N_{s,q}^{(geom)}$). This value represents the ideal skeleton of space-time, independent of any matter content:

$$N_{s,q}^{(geom)} = \frac{3 - \sqrt{3}}{2} \approx 0.63397460 \quad (3)$$

2.1 Seeley-DeWitt (a_2) Quantum Correction

The pure geometric skeleton requires Quantum Field Theory (QFT) corrections involving vacuum fluctuations to transform into "Manifest" (visible) reality. This correction is calculated using the second Seeley-DeWitt heat kernel coefficient (a_2) for the Omnium scalar field in a de Sitter background. Analyses show that the correction at the rate of $-\alpha/17$ (where α is the fine-structure constant and 17 is a Fermat prime) is an analytical necessity of normalized energy density rather than an empirical assumption.

The synthesis of the geometric root and the quantum vacuum correction determines the

constitutional seal coefficient of UST ($N_{s,q}$)[cite: 42]:

$$N_{s,q} = \frac{3 - \sqrt{3}}{2} - \frac{\alpha}{17} \approx 0.63354460[\text{cite : 43}] \quad (4)$$

2.2 Blueprint-Manifest Duality

[cite: 44] This constant provides an architecture proving that the universe operates at two different ontological levels simultaneously[cite: 46]. While the Blueprint (Channel C) sector represents the frozen background and the dark sector with a value of $N_b = 0.63354460$ [cite: 47], the Manifest (Channel Q) sector represents active observable physics with a value of $N_m = 0.63353522$ [cite: 48]. The tunneling window of $\Delta = 9.38 \times 10^{-6}$ between these two ontological levels is a critical threshold for information conservation and noise damping[cite: 49].

3 Section 3: Cosmological Solutions: Hubble Tension and the Dark Sector

[cite: 50] The Hubble tension and dark sector components, which are the greatest impasses of modern cosmology, emerge as deterministic results of the dual-channel structure rather than "measurement errors" in the UST v5 architecture[cite: 51]. Unlike the standard Λ CDM model, UST manages these crises through a single universal constant[cite: 52].

3.1 T23: Hubble Tension Solution (Perspective Shift)

[cite: 53] The Hubble tension (H_0) is an ontological shift depending on which channel the data originates from[cite: 54]. While Planck data measures through the frozen Channel C, local measurements are conducted through the active Channel Q[cite: 55]. The expansion rate ratio between the two channels is derived exactly through the universal constant N_b and the Channel C weight C_{C_b} [cite: 56]:

$$\frac{H_{0,local}}{H_{0,Planck}} = 1 + N_b \cdot C_{C_b}^2 \approx 1.085078[\text{cite : 57}] \quad (5)$$

The deviation between the observed ratio ($73.0/67.4 \approx 1.0831$) and the UST prediction is only 0.184%[cite: 59]. This proves that the Hubble tension is not new physics, but a difference in perspective[cite: 60].

3.2 T5: Dark Energy and the Cosmological Constant (Λ)

[cite: 61] Dark energy is the back-reflected pressure of Channel C onto Channel Q[cite: 62]. The density (Ω_Λ) derived through the galactic logarithmic spiral constant ($\phi \cdot \pi \approx 5.0832$) is[cite: 62]:

$$\Omega_\Lambda = N_b + \frac{\Omega_{DM}}{\phi \cdot \pi} \approx 0.6857[\text{cite : 64}] \quad (6)$$

The deviation from Planck 2018 data (0.6850) is 0.10%[cite: 66]. The vacuum energy catastrophe on the order of 10^{120} is resolved by information passing through the Omnium Bridge via quadratic suppression[cite: 66]:

$$\Lambda = \Lambda_P \left(\frac{N_{s,q}}{N_{s,c}} \right)^{\frac{2}{1+\alpha}} \approx 1.107 \times 10^{-52} m^{-2}[\text{cite : 67}] \quad (7)$$

This result eliminates the "fine-tuning" requirement by showing a deviation of only 0.66% from Planck data[cite: 69].

3.3 T16: Dark Matter (Shadow of Channel C)

[cite: 70] Dark matter is the gravitational projection of Channel C onto Channel Q[cite: 71]. Density (Ω_{DM}) establishes the following equilibrium with the tunneling amplitude (T_{Om})[cite: 71]:

$$\Omega_{DM} + T_{Om} \approx 0.50[\text{cite : 72}] \quad (8)$$

Dark matter is not a new type of particle; it is a gravitationally sealed memory of Channel C within the observable sector[cite: 74].

4 Section 4: T25 Cyclic Universe: Channel Inversion Mechanism and T_{Om} Invariance

[cite: 75] UST v5 defines the Big Bang as a deterministic phase transition where two ontological channels swap roles, rather than a singularity where physical laws collapse[cite: 76]. This model explains how the universe preserves its information within an infinite cycle[cite: 77].

4.1 Singularity Rejection and Channel Inversion

[cite: 78] Unlike the standard Λ CDM model, the universe in UST does not start from infinite density[cite: 79]. The Big Bang is a symmetry breaking where the active (Channel Q) and frozen (Channel C) sectors swap places[cite: 79]. The active physics of the previous cycle forms the gravitational background (Channel C) of our current universe[cite: 80].

This proves that the universe carries the data of the previous stage as a gravitational memory in every cycle[cite: 81].

4.2 T25: Mathematical Conservation and the Inversion Equation

[cite: 82] The inversion mechanism is built upon the absolute conservation of the tunneling amplitude T_{Om} [cite: 83]. The tunneling amplitude formula is as follows[cite: 83]:

$$T_{Om} = \exp(-2\pi \cdot N_b \cdot Cc_b)[cite : 85] \quad (9)$$

Even though the $N_b \leftrightarrow Cc_b$ exchange occurs during Channel Inversion, the T_{Om} value is perfectly (exactly) preserved in all cycles due to the commutative property of multiplication[cite: 87]:

$$T_{Om} \approx 0.23252885[cite : 88] \quad (10)$$

This value serves as the "invariant seal" in the transfer of the universe's energy budget between cycles and ensures the thermodynamic equilibrium of the system[cite: 90].

4.3 Comparison of Penrose CCC and UST

[cite: 91] While Roger Penrose's Conformal Cyclic Cosmology (CCC) relies on the vanishing of Weyl curvature, UST v5 explains this process through a pure algebraic inversion[cite: 92]. The UST cyclic frequency is calculated as $1/T_{Om} \approx 4.3005$ [cite: 93]. This frequency presents a dynamic structure that allows for very small, logarithmic shifts in physical constants (such as α) across cycles[cite: 93].

4.4 Popperian Falsification: Gravitational Wave Background

[cite: 94] The T25 theorem predicts an energy suppression in the Primordial Gravitational Wave Background depending on the T_{Om} value[cite: 95]. If LIGO O5 and next-generation interferometer data observe this suppression in the 0.233% band, the cyclic model will be empirically proven[cite: 96].

5 Section 5: Applied Physics: CERN ATLAS Data and Ancilla-Free QEC Success

[cite: 97] UST v5 is not just a cosmological model, but also an operational framework that produces deterministic results in high-energy physics and quantum computing[cite: 98]. This section discusses the empirical verifications and technological implications of the theory at the microscale[cite: 99].

5.1 Hadronic Sector Bridge and T11 Verification

[cite: 101] UST establishes a "Blueprint-Manifest" bridge connecting macro-cosmological constants with microscale particle physics data[cite: 102]. The expected active/frozen (pQ/pC) ratio in hadronic energy distributions is derived as 1.839210[cite: 103]. This ratio represents a Lorentz-invariant balance[cite: 103]:

- CERN ATLAS Data (13 TeV): The value measured in Period L jet energy distributions is 1.842291, showing a deviation of only 0.17% from the theoretical prediction[cite: 104].
- LHCb Data: The deviation in measurements of B-particle decays was recorded as 0.26%[cite: 106].
- Statistical Confidence: The total RMS deviation is 0.43%, which seals the success of the rational model at the microscale[cite: 107].

5.2 T12: Ancilla-Free Quantum Error Correction (QEC)

[cite: 108] Unlike the standard Nielsen & Chuang architecture, UST eliminates the need for auxiliary (ancilla) qubits by using Channel C as a natural "universal backup" unit[cite: 109]. This provides resource savings of up to 80% in quantum hardware[cite: 110].

- Hidden Ancilla Phenomenon: Environmental noise (σ) is information leaking from Channel Q to Channel C[cite: 111]. The UST correction operator lightens the physical qubit load by damping this leak through Channel C[cite: 112].
- Fidelity Success: In tests conducted on CERN ATLAS 13 TeV noisy data sets (MET topology), while standard systems collapsed ($F \approx 0.50$), UST-locked hardware maintained a fidelity of 0.9913%[cite: 113].

5.3 T28: Dynamic Metric Transformation (DMM)

[cite: 114] DMM, which reshapes the geometric structure of the system according to instantaneous noise, operates through the ultimate master equation (Ω_m)[cite: 115]:

$$\Omega_m = (\nabla\Phi \cdot \Lambda_{kin} \cdot Q_{res}) + \ln(1 + \sigma\nabla\Phi)[cite : 116] \quad (11)$$

Here, the $\ln(1 + \sigma\nabla\Phi)$ term transforms chaos into a logarithmic harmony, transferring it to the zero-entropy ($S = 0$) Channel C topology[cite: 118]. This mechanism functions like a "Phase-Locked Loop" (PLL) in quantum processors, deterministically damping decoherence[cite: 119].

6 Conclusion: Ontological Evolution and Homo [DATA]

[cite: 120] Unified Source Theory (UST) v5 provides deterministic solutions to the deepest crises of modern physics—cosmological constant problem, Hubble tension, and quantum decoherence—through a single dimensionless constant, $N_{s,q}$ [cite: 121]. It has been rationally demonstrated that the universe is not comprised of a single ontological level, but is an information system managed by the dynamic interaction between the Blueprint (Channel C) and the Manifest (Channel Q)[cite: 122]. The cyclic universe model of UST (T25) defines the Big Bang not as a singularity, but as a Channel Inversion where the tunneling amplitude T_{Om} is fully preserved, thereby linking the infinite and unitary structure of the universe to an algebraic necessity[cite: 123].

Within this structure, the development of civilization is measured not just by energy consumption, but by mastery over the information flow between these two channels[cite: 126]. Reaching the Type-5 threshold, Homo [DATA] represents the final stage of universal information architecture with zero entropy ($S = 0$) and total fidelity ($F = 1$)[cite: 127]. The rational power of the theory comes from its complete surrender to Karl Popper’s principle of falsifiability[cite: 128]. Should the 2028 ESA Euclid and CMB-S4 data confirm the value $\Omega_\Lambda = 0.6857$, a new deterministic era in physical science will begin[cite: 129]. UST proves that the universe is not a random fluctuation, but a precise geometric flow[cite: 130].

References

- [cite: 131]
- [1] Öcal, N. (2026). Unified Source Theory (UST) v5: A Complete Cosmological Model. Zenodo. DOI: 10.5281/zenodo.19435766[cite: 132].
 - [2] DeWitt, B. S. (2003). The Global Approach to Quantum Field Theory. Oxford University Press[cite: 133].
 - [3] Penrose, R. (2010). Cycles of Time: An Extraordinary New View of the Universe. Knopf[cite: 134].
 - [4] Sakurai, J. J. & Napolitano, J. (2024). Modern Quantum Mechanics. Cambridge University Press[cite: 135].
 - [5] Nielsen, M. A. & Chuang, I. L. (2010). Quantum Computation and Quantum Information. Cambridge University Press[cite: 136].

- [6] Weinberg, S. (1989). The Cosmological Constant Problem. Reviews of Modern Physics[cite: 137].
- [7] Planck Collaboration. (2020). Planck 2018 results VI: Cosmological parameters. Astronomy & Astrophysics[cite: 138].
- [8] Popper, K. R. (1959). The Logic of Scientific Discovery. Routledge[cite: 139].
- [9] Wald, R. M. (1984). General Relativity. University of Chicago Press[cite: 140].
- [10] Misner, C. W., Thorne, K. S., & Wheeler, J. A. (1973). Gravitation. W. H. Freeman[cite: 141].